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MTH 240 Midterm 2

Q1 8

8 Marks

1. Determine whether each of the following series is convergent or divergent. If it is convergent find its sum.

(a) $\sum_{n=1}^{\infty} \ln\left(\frac{n^2+1}{5n^2+3}\right)$

Test for Divergence:

$\lim_{n\to\infty} \ln\left(\frac{n^2+1}{5n^2+3}\right)$

$= \ln\left(\lim_{n\to\infty} \frac{n^2+1}{5n^2+3}\right)$

$= \ln\left(\lim_{n\to\infty} \frac{\frac{n^2}{n^2}+\frac{1}{n^2}}{\frac{5n^2}{n^2}+\frac{3}{n^2}}\right)$

$= \ln\left(\frac{1}{5}\right) = -\ln 5 \neq 0$

$\therefore$ By Test for Divergence, $\sum_{n=1}^{\infty} \ln\left(\frac{n^2+1}{5n^2+3}\right)$ diverges.

(b) $\sum_{n=1}^{\infty} \frac{6 \cdot 2^{2n-1}}{5^{n+1}}$

$= 6 \sum_{n=1}^{\infty} \frac{4^n\left(\frac{1}{2}\right)}{5^n \cdot 5}$

$= \frac{3}{5} \sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^n$

$= \left(\frac{3}{5}\right)\left(\frac{4}{5}\right) \sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^{n-1}$

Since $r = \frac{4}{5} < 1$, this series converges

The sum:

$= \frac{12/25}{1-4/5}$

$= \frac{12/25}{1/5}$

$= \frac{12}{5}$

$\therefore \sum_{n=1}^{\infty} \frac{6 \cdot 2^{2n-1}}{5^{n+1}} = \frac{12}{5}$,

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Q2 8

8 Marks

2. Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{(1+4n^2)^{3/2}}$$

(a) Explain why the integral test can be applied to this series.

This is because it satisfies the three necessary conditions:

1) for all n, this series is positive (numerator and denominator both always positive)

2) for all n, this series is continuous $(1+4n^2) \neq 0$ for all n

3) If the series were a function, $f(x) = (1+4x^2)^{-3/2}$,

$f'(x) = -\frac{3}{2}(1+4x^2)^{-5/2}(8x)$, which is always negative, $\therefore f(x)$ and thus the series decrease for $n \in [1, \infty)$.

(b) Use the integral test to determine whether this series is convergent or divergent.
Note: No credit will be given if a different method is used.

$$\int_1^{\infty} \frac{1}{(1+4x^2)^{3/2}}\,dx$$

$$= \lim_{t\to\infty} \int_1^t \frac{1}{(1+4x^2)^{3/2}}\,dx$$

sub $x = \frac{1}{2}\tan\theta$

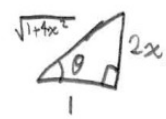
$dx = \frac{1}{2}\sec^2\theta\, d\theta$

$$= \lim_{t\to\infty} \int_{x=1}^{x=t} \frac{\frac{1}{2}\sec^2\theta}{(1+\tan^2\theta)^{3/2}}\,d\theta$$

$$= \lim_{t\to\infty} \int_{x=1}^{x=t} \frac{\frac{1}{2}\cancel{\sec^2\theta}}{\sec^{\cancel{3}}\theta}\,d\theta$$

$$= \frac{1}{2}\lim_{t\to\infty} \int_{x=1}^{x=t} \cos\theta\, d\theta$$

$$= \frac{1}{2}\lim_{t\to\infty} \sin\theta \Big|_{x=1}^{x=t}$$



$$= \frac{1}{2}\lim_{t\to\infty} \left(\frac{2x}{\sqrt{1+4x^2}}\right)\Big|_1^t$$

$$= \frac{1}{2}\lim_{t\to\infty} \left[\frac{2t}{\sqrt{1+4t^2}} - \frac{2}{\sqrt{5}}\right]$$

$$= \frac{1}{2}\left[\lim_{t\to\infty} \frac{2}{\sqrt{\frac{1}{t^2}+4}} - \frac{2}{\sqrt{5}}\right]$$

$$= \frac{1}{2}\left[1 - \frac{2}{\sqrt{5}}\right]$$

$$= \frac{1}{2} - \frac{1}{\sqrt{5}}$$

Since $\int_1^{\infty} \frac{1}{(1+4x^2)^{3/2}}\,dx$ converges

$\sum_{n=1}^{\infty} \frac{1}{(1+4n^2)^{3/2}}$ converges by Integral Test.

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8 Marks

3. Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

Q3 8

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\sqrt{2n^2-1}}{5n^2-3n+1}$$

$$a_n = \frac{(-1)^{n-1}\sqrt{2n^2-1}}{5n^2-3n+1}$$

Alternating Series Test:

$$|a_n| \Rightarrow f(x) = \frac{\sqrt{2x^2-1}}{5x^2-3x+1}$$

$$f'(x) = \frac{\frac{2x}{\sqrt{2x^2-1}}(5x^2-3x+1) - \sqrt{2x^2-1}\,(10x-3)}{(5x^2-3x+1)^2}$$

Including $\sqrt{2x^2-1}$ (when numerator made into single fraction).

$\Rightarrow$ Ignore denominators because their always positive.

$10x^3 - \cancel{6x^2} + 2x - 20x^3 + \cancel{6x^2} + 10x - 3 \leq 0$

$-10x^3 + 12x - 3 \leq 0$ eventually since highest degree has negative coefficient.

$\therefore$ Alternating Series Test applies

$$\lim_{n\to\infty} \frac{\sqrt{2n^2-1}}{5n^2-3n+1} \cdot \frac{1/n^2}{1/n^2} = \lim_{n\to\infty} \frac{\sqrt{\frac{2}{n^2} - 1/n^4}}{5 - \frac{3}{n} + \frac{1}{n^2}} = 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ converges by Alternating Series Test.

For $b_n = |a_n|$, use Limit Comparison Test:

Choose $c_n = \frac{n}{n^2} = \frac{1}{n}$

$$\lim_{n\to\infty} \frac{b_n}{c_n} = \lim_{n\to\infty} \frac{\sqrt{2n^2-1}}{5n^2-3n+1} \cdot \frac{1/n}{1/n^2} = \lim_{n\to\infty} \frac{\sqrt{\frac{2n^2}{n^2} - \frac{1}{n^2}}}{5 - \frac{3}{n} + \frac{1}{n^2}} = \frac{\sqrt{2}}{5} > 0 \text{ and finite.}$$

$\therefore$ Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges (harmonic series), $\sum_{n=1}^{\infty} \frac{\sqrt{2n^2-1}}{5n^2-3n+1}$ diverges also.

$\therefore$ Since $\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} |a_n|$ diverges, $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\sqrt{2n^2-1}}{5n^2-3n+1}$ is

conditionally convergent.

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Q4 8

8 Marks

4. Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{(2n)!}{n^{n-1}(n+1)^{n+1}}$$

Ratio Test:

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(2n+2)!}{(n+1)^n(n+2)^{n+2}} \cdot \frac{n^{n-1}(n+1)^{n+1}}{(2n)!}\right|$$

$$= \lim_{n\to\infty}\left|\frac{(2n+2)(2n+1)\cancel{(2n)!}}{\cancel{(n+1)^n}(n+2)^n(n+2)^2} \cdot \frac{n^n\cancel{(n+1)^n}(n+1)}{n\cancel{(2n)!}}\right|$$

$$= \lim_{n\to\infty}\left|\frac{(2n+2)(2n+1)(n+1)n^n}{(n+2)^n(n+2)^2 n}\right|$$

$$= \lim_{n\to\infty}\left|\frac{(2n+2)(2n+1)(n+1)}{n\cdot(n+2)^2} \cdot \left(\frac{n}{n+2}\right)^n\right|$$

$$= \lim_{n\to\infty}\left|\frac{(4n^2+6n+2)(n+1)}{n^3+4n^2+4n} \cdot \left(\frac{1}{1+\frac{2}{n}}\right)^n\right|$$

$$= \lim_{n\to\infty}\left|\frac{4n^3+6n^2+2n+4n^2+6n+2}{n^3+4n^2+4n} \cdot \left(\frac{1}{1+\frac{2}{n}}\right)^n\right|$$

$$= \lim_{n\to\infty}\left|\frac{4n^3+10n^2+8n+2}{n^3+4n^2+4n}\right| \cdot \lim_{n\to\infty}\left|\left(\frac{1}{1+\frac{2}{n}}\right)^n\right|$$

$$= \lim_{n\to\infty}\left|\frac{4+\frac{10}{n}+\frac{8}{n^2}+\frac{2}{n^3}}{1+\frac{4}{n}+\frac{4}{n^2}}\right| \cdot e^{-2}$$

$$= \frac{4}{e^2} < 1$$

$\therefore$ By Ratio Test, $\sum_{n=1}^{\infty} \frac{(2n)!}{n^{n-1}(n+1)^{n+1}}$ converges.

MTH 240 Midterm 2

Q5 8

8 Marks

5. Solve the differential equation (do not solve for y):

$$4y\frac{dy}{dx} = (y^4 - 1)\tan^{-1}(x)$$

$$\frac{4y}{y^4-1}\,dy = \tan^{-1}x\,dx$$

$$\int \frac{4y}{y^4-1}\,dy = \int \tan^{-1}x\,dx$$

Sub $u = y^2$

$du = 2y\,dy$

$2\int \frac{1}{u^2-1}\,du = *$

Partial Fractions:

$\frac{A}{u+1} + \frac{B}{u-1} = \frac{1}{u^2-1}$

$A(u-1) + B(u+1) = 1$

$u=-1,\ A = -\frac{1}{2}$

$u=1,\ B = \frac{1}{2}$

$* = 2\int \frac{-1/2}{u+1} + \frac{1/2}{u-1}\,du$

$= \int \frac{1}{u-1} - \frac{1}{u+1}\,du$

$= \ln|u-1| - \ln|u+1|$

$= \ln\left|\frac{u-1}{u+1}\right|$

$= \ln\left|\frac{y^2-1}{y^2+1}\right| + C$

$u = \tan^{-1}x \quad du = \frac{1}{x^2+1}$

$dv = dx \quad v = x$

$= x\tan^{-1}x - \int \frac{x}{x^2+1}\,dx$

$u = x^2+1$

$du = 2x\,dx$

$= x\tan^{-1}x - \frac{1}{2}\int \frac{1}{u}\,du$

$= x\tan^{-1}x - \frac{1}{2}\ln|x^2+1| + C$

$\therefore \ln\left|\frac{y^2-1}{y^2+1}\right| = x\tan^{-1}x - \frac{1}{2}\ln|x^2+1| + C$

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MTH 240 Midterm 2

10 Marks 6. Solve the initial value problem

Q6 10

$$\frac{1}{\sqrt{x}}\frac{dy}{dx} = \frac{x^{3/2}(1-12y)}{x^3+2} \qquad \text{where} \qquad y(1)=1$$

$(x^3+2)\frac{dy}{dx} = (x^{3/2} - x^{3/2}(12y))\sqrt{x}$

$(x^3+2)\frac{dy}{dx} = x^2 - 12x^2y$

$(x^3+2)\frac{dy}{dx} + 12x^2y = x^2$

$\frac{dy}{dx} + 12\frac{x^2}{x^3+2}y = \frac{x^2}{x^3+2}$

$I(x) = e^{4\int \frac{3x^2}{x^3+2}dx} = e^{4\ln|x^3+2|}$

$\therefore I(x) = (x^3+2)^4$

$(x^3+2)^4\frac{dy}{dx} + 12(x^3+2)^3x^2y = x^2(x^3+2)^3$

$(y(x^3+2)^4)' = x^2(x^3+2)^3$

$y(x^3+2)^4 = \int x^2(x^3+2)^3\,dx$ Sub $u = x^3+2$, $du = 3x^2dx$

$y(x^3+2)^4 = \frac{1}{3}\int u^3\,du$

$y(x^3+2)^4 = \frac{1}{3}\frac{u^4}{4} + C$

$y(x^3+2)^4 = \frac{(x^3+2)^4}{12} + C$

At $y(1)=1$,

$(1)(1^3+2)^4 = \frac{(1^3+2)^4}{12} + C$

$3^4 = \frac{3^4}{12} + C$

$\frac{3^4(12)-3^4}{12} = C$

$\frac{3^4(12-1)}{12} = C$

$\frac{81(11)}{12} = C$

$C = \frac{891}{12}$

$y(x^3+2)^4 = \frac{(x^3+2)^4}{12} + \frac{891}{12}$

$\therefore y = \frac{1}{12}\left(1 + \frac{891}{(x^3+2)^4}\right)$

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Space2